BUSINESS INTELLIGENCE LAB

EXPERIMENT 5: REPORT GENERATION USING RELATIONAL AND MULTIDIMENSIONAL DATA

1. Problem Statement

Generate interactive reports and a dashboard in Microsoft Power BI using the Global Superstore dataset. The reports must demonstrate both relational (tabular) and multidimensional (hierarchical and aggregated) views of the data. The dashboard should help users analyze sales, profit, orders, customers, products, and geographical performance.

2. Aim

To generate relational and multidimensional reports using Microsoft Power BI and analyze business performance through tables, hierarchies, charts, KPI cards, and slicers.

3. Objectives

- Import the Global Superstore Dataset.xlsx file into Power BI Desktop.
- Prepare and inspect the data using Power Query Editor.
- Create a relational table containing important order and sales fields.
- Build a Region–State–City hierarchy and analyze data using a Matrix visual.
- Create charts for category-wise sales, region-wise profit, top customers, and year-wise sales trends.
- Design an interactive dashboard using KPI cards and slicers.
- Interpret the generated reports and identify useful business insights.

4. Software and Tools Required

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- Data Analysis Expressions (DAX), where required

5. Expected Output

- A relational table displaying selected order-level details.
- A Matrix report showing sales, profit, and quantity across Region, State, and City.
- Four analytical visuals showing category, regional, customer, and yearly performance.
- An interactive dashboard containing three KPI cards and three slicers.

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6. Result

The experiment is considered successfully completed when the relational table, hierarchical Matrix, required analytical visuals, and interactive dashboard are created from the Global Superstore dataset in Microsoft Power BI.